



DESIGN PROPOSAL · 01

Mobile experience redesign

A screen-by-screen proposal for how ResearchAI should look and behave on a phone — replacing the shrunk-desktop experience with a native-feeling mobile app pattern, in the product's existing white-and-green language.

Landing

Sign in

Register

Dashboard

Generate wizard

Projects

Spec reader

Topic Lab

Profile

The mobile navigation model

TODAY

The desktop layout simply collapses: the 232px sidebar disappears below 860px and all navigation moves behind a **hamburger + slide-out drawer**. Every destination costs two taps, current location is invisible, tap targets are desktop-sized (~32px), tables and multi-column grids overflow, and page headers scroll away.

PROPOSED

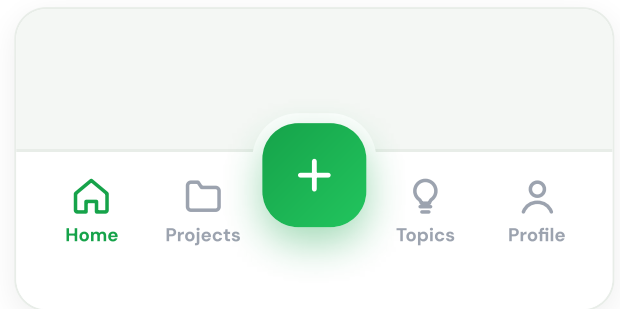
A persistent **bottom tab bar** — the four destinations that matter, always one thumb-tap away, with the app's single most important action (**New Spec**) as a raised center button:

- 1 **Home** — dashboard overview & recent activity
- 2 **Projects** — every generated specification
- 3 **New Spec** — center FAB, opens the Generate wizard
- 4 **Topics** — the Topic Lab toolset
- 5 **Profile** — account, API key, model tiers

WHY A TAB BAR, NOT THE DRAWER

This app has exactly 5 destinations — the textbook case for bottom tabs (Apple HIG & Material both). Navigation is always visible, state is self-evident, and reach is thumb-native. The drawer pattern exists for apps with 7+ sections; here it only hides the product.

TAB BAR ANATOMY

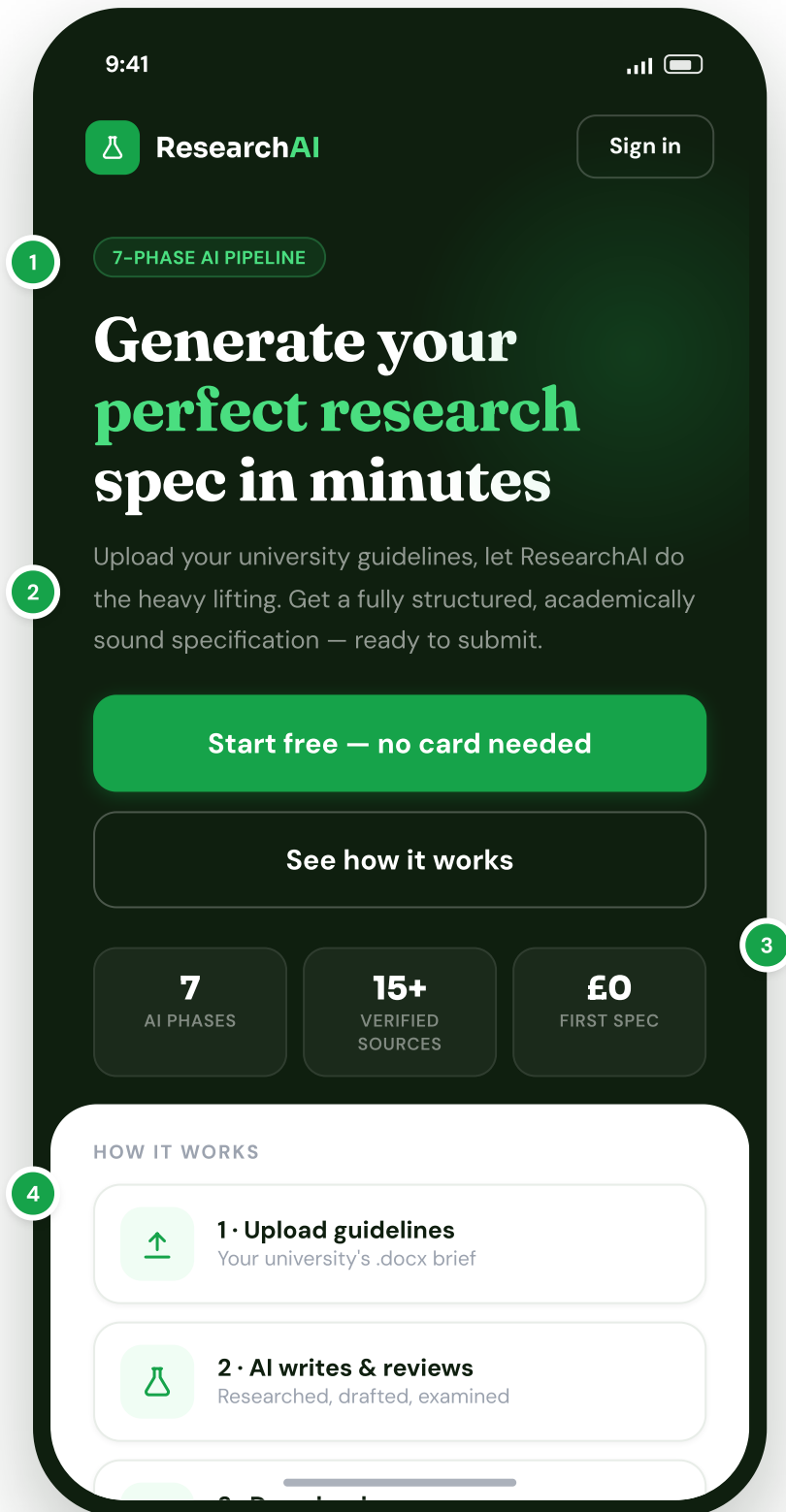


- a Active tab in **brand green**, inactive in muted gray — never two greens competing.
- b FAB raised 22px above the bar, 52×52px, green gradient + white ring — same treatment as the desktop "Generate Spec" card.
- c Bar floats above the iOS home indicator (safe-area inset).

GLOBAL RULES

- Touch targets ≥ **44×44px**, list rows ≥ 56px
- Inputs ≥ 16px font — prevents iOS auto-zoom
- One column everywhere; tables become cards
- Sticky app bar: page title + avatar, 56px
- Sticky bottom CTA on forms (never below the fold)
- Type: Sora headings · DM Sans UI · Fraunces/serif for document reading
- Existing tokens unchanged: **#16a34a** on white, **#0f10f** ink

Landing



- 1 Compact nav, one action.** Logo + a single "Sign in" pill. The full menu lives in the page flow, not a drawer — a marketing page has no business hiding its content.
- 2 The real headline, resized — not rewritten.** Fraunces at 2rem/1.12 with the green *em* kept. Both CTAs full-width and stacked; primary carries the free-trial promise ("no card needed") since the free credit is now a real product feature.
- 3 Stats as a 3-up strip** instead of the desktop 4-column band — glanceable without scrolling past a wall.
- 4 White sheet rises over the dark hero** (24px top radius) — a deliberate "app-like" seam that carries the dark→light transition the desktop page does with sections. "How it works" becomes 3 tappable rows.

KEEPS THE THEME

Dark forest (#0f1f0f) hero with green glow + dot grid is lifted straight from the auth pages' left panel — the strongest visual asset the product already owns. Below the fold: features single-column, pricing cards stacked (Free / £9.99 / £24.99 with "Most Popular" on the middle card), then the dark CTA band.

Sign in

9:41

 **ResearchAI**

Sign in

Welcome back — pick up where you left off.

 **Continue with Google**

or with email

EMAIL

jane@university.ac.uk

PASSWORD [Forgot?](#)

.....

Sign in

New here? [Create an account](#)

 Your first specification is **free** — no card, no API key needed.

- 1 Brand header replaces the lost panel.**
Today the dark left panel simply vanishes under 900px, taking all brand presence with it. The logo + green accent bar restores identity in 60px of height.
- 2 Google first, full-width, 50px tall.** It's the lowest-friction path and now works in production — give it the top slot. Real GSI button sits invisibly on top of the styled one (the pattern the app already uses).
- 3 "Forgot?" moves inline with the password label** — one less row, and it can't be missed at the exact moment it's needed.
- 4 The free-trial promise fills the dead space** where desktop shows proof cards — it's the one message most likely to convert a hesitant student, anchored safely above the home indicator.

STATES COVERED

Error banner slides under the app bar (red tint, matches .bg-r badge). "Email unverified" state swaps the primary button for "Resend verification" with a 60s countdown — same components, no new patterns. All inputs 48px+, 16px text: no iOS zoom-jump.

Create your account

9:41

 ResearchAI

Create your account

Your first spec is on us.

 Sign up with Google

or

FULL NAME

Jane Smith

EMAIL

jane@university.ac.uk

PASSWORD

.....

Good

☒ I agree to the [Terms](#) and [Privacy Policy](#)

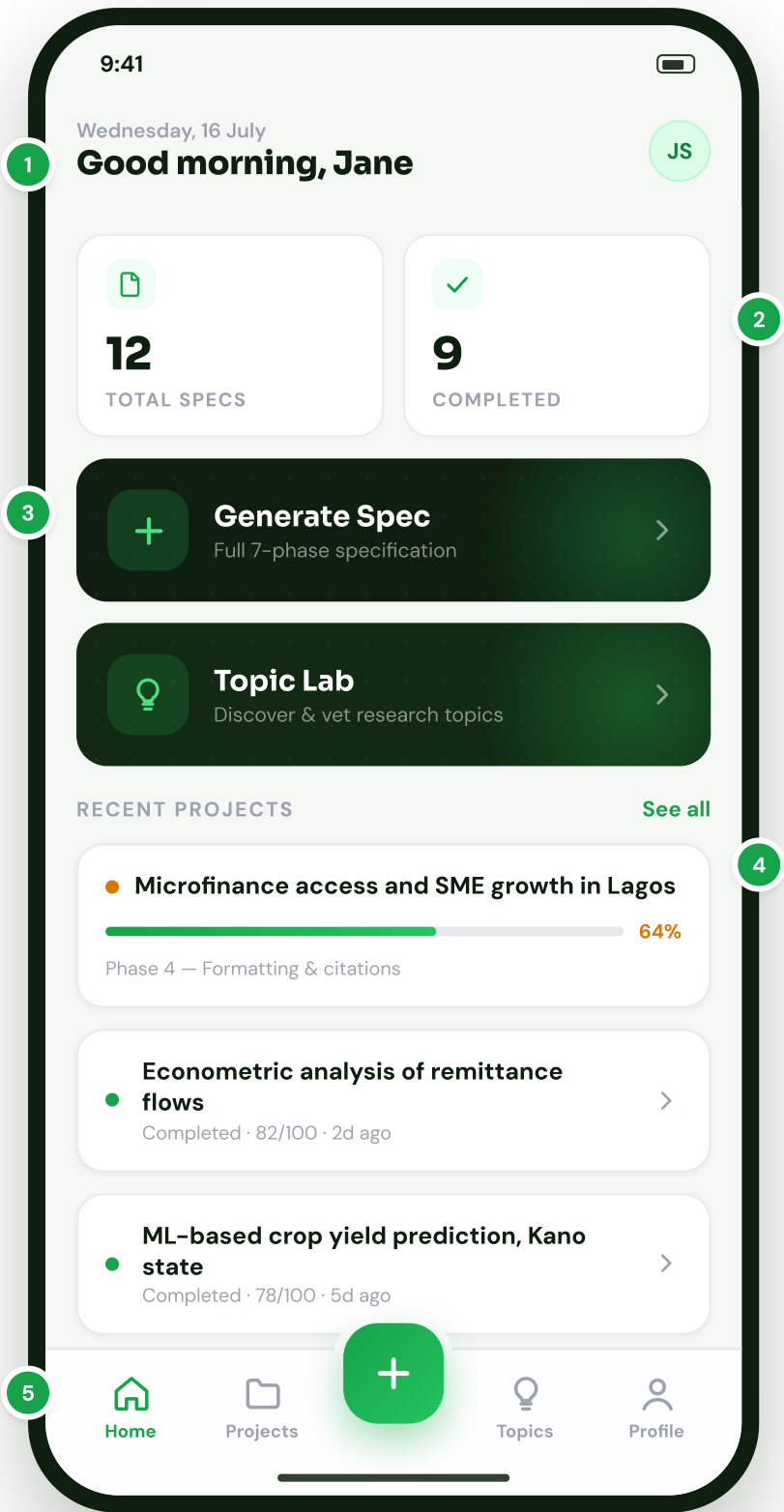
Create account

Already registered? [Sign in](#)

- 1 **Live password-strength meter** — four segments filling in green, replacing silent validation. Costs one div, saves a round-trip failure.
- 2 **44px checkbox row** for terms — the tap area is the whole row, not the 16px box.
- 3 **Sticky bottom CTA.** "Create account" is pinned above the home indicator, visible even with the keyboard open (the form scrolls, the button doesn't).

POST-SUBMIT: "ACCOUNT CREATED!"

The existing success state becomes a full-screen moment: green check-circle animation, "Account created!" in Sora 800, "We've sent a verification link to jane@..." and a single "Open mail app" button (mailto: deep-link). The resend option appears after 30s. Same flow as today, one screen instead of a squeezed card.

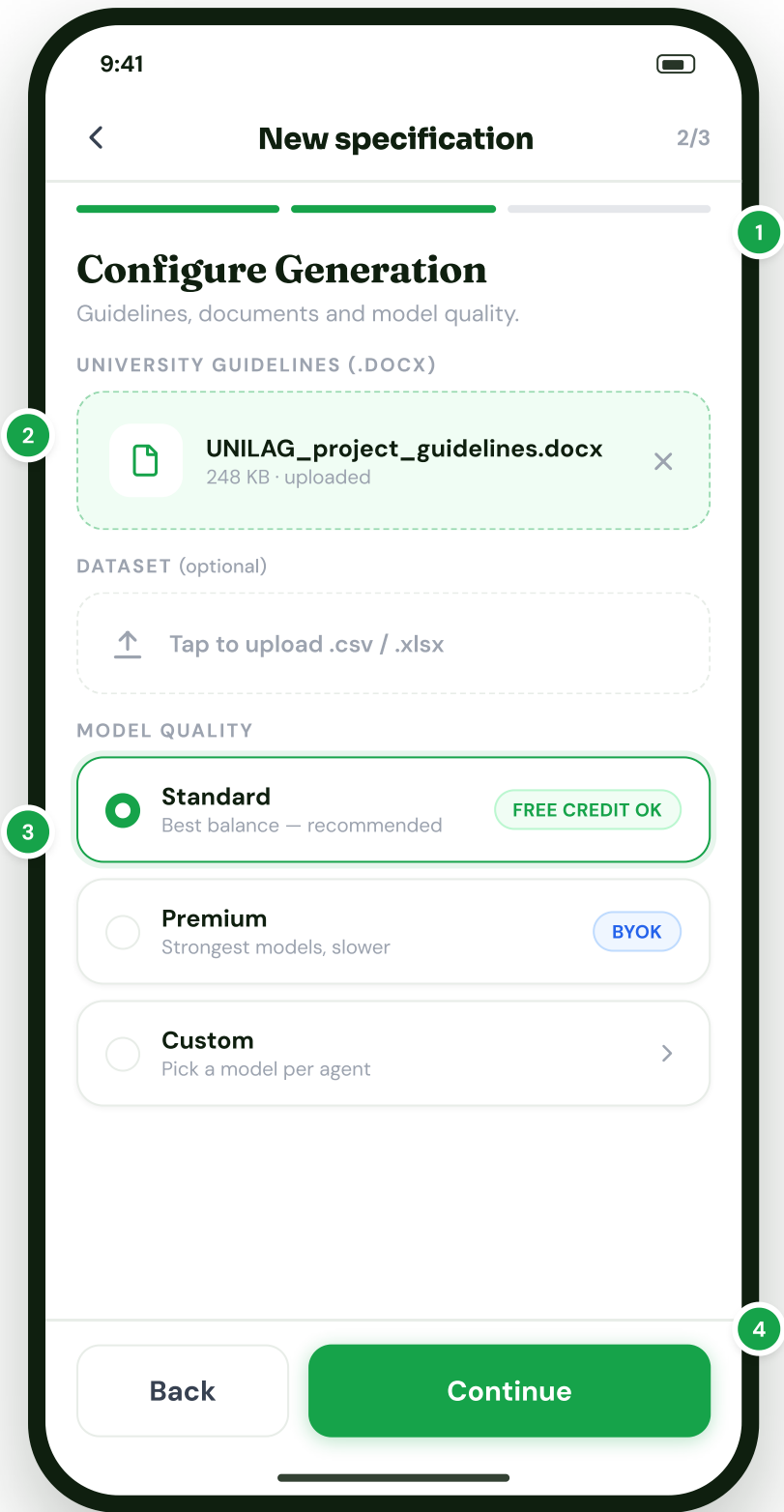


- 1 **Greeting header** with date — warmer than a bare "Dashboard" title, and the avatar doubles as a shortcut to Profile.
- 2 **Two stat tiles, not four.** Hero numbers in ink (not green — color is for accents, numbers are for reading), labels muted uppercase. "In progress" and trial credits live lower down where they have context.
- 3 **The two dark action cards survive** — they're the strongest element of the current dashboard — resized to full-width rows with the dot-grid + glow treatment kept.
- 4 **A running generation gets a live card:** amber pulse dot, progress bar, and the real phase description ("Phase 4 — Formatting & citations") the API already returns. Polling already exists; this just surfaces it on Home.
- 5 **Completed rows carry their mark** (82/100) — the single number users care about — with the green status dot (the one this session's typo fix made work again).

WHAT DISAPPEARS

The desktop's table layout, the sidebar usage meter (moves to Profile), and all hover-dependent affordances. Everything above is reachable with one thumb; nothing needs a tooltip to be understood.

Generate — a true mobile wizard



- 1 **One step per screen, segmented progress** + "2/3" count. The desktop page stacks Basic Information, Configure Generation and Review & Generate into one long scroll — on mobile each becomes its own screen with a single job.
- 2 **Upload zones become tap targets**, not drag-and-drop wells. Uploaded state shows filename + size + remove; the dashed green well is the app's existing visual, kept.
- 3 **Model tiers as radio cards** with the cost story on each: Standard is free-credit eligible, Premium flags BYOK. This is where the free-trial/BYOK feature becomes self-explanatory instead of a surprise 402.
- 4 **Sticky footer nav** — Back never competes with Continue (110px vs. flexible), and both clear the home indicator.

THE THREE STEPS

1 ·

BASICS

— field of study, topic, academic level (large select sheets, not dropdowns)

2 ·

CONFIGURE

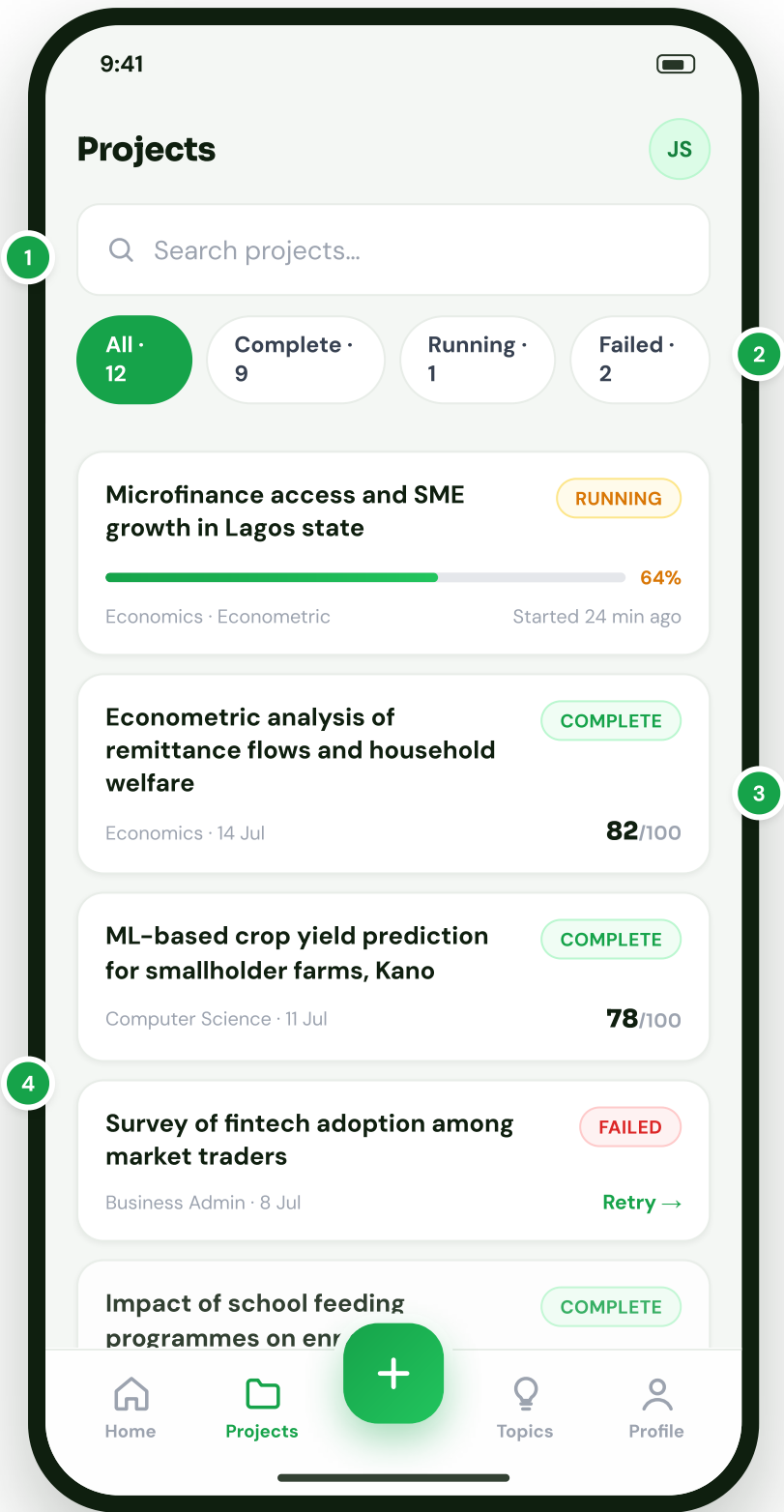
— this screen

3 ·

REVIEW & GENERATE

— summary card of every choice, estimated iterations (max_iterations 1–5 as a stepper), then a full-width green "Generate specification". Submitting routes straight to the live progress screen (sheet 10).

Projects

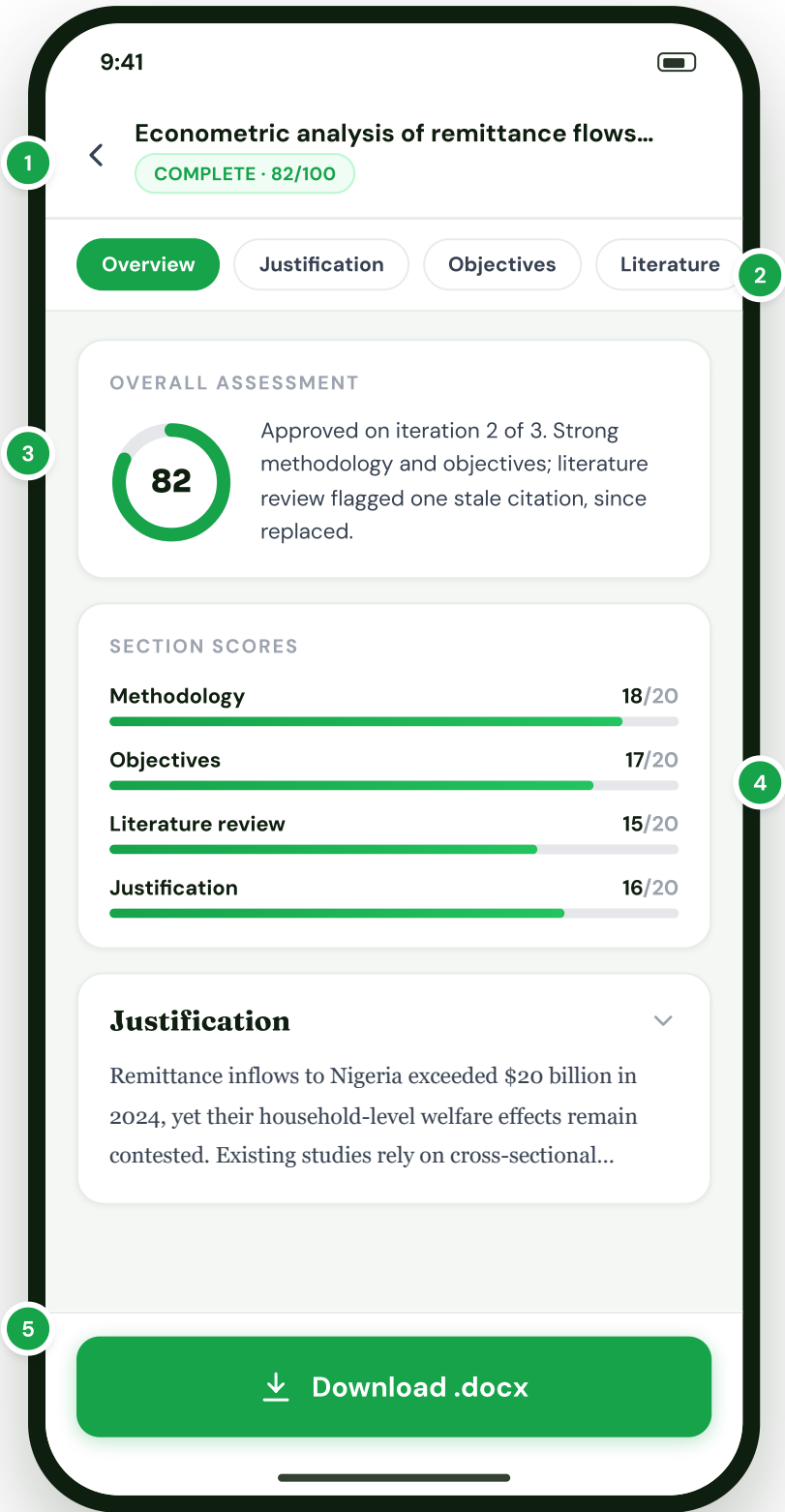


- 1 **Search + filter chips replace the desktop table's controls.** Chips carry counts so the filter is informative before it's tapped. Invalid statuses can't be expressed — the chips *are* the enum.
- 2 **Status as a badge, top-right, always visible** — the desktop's colored-dot column doesn't survive a 360px row; badges + the title's two-line clamp do.
- 3 **Completed cards lead with the examiner mark** in Sora 800 — the number students actually compare across projects.
- 4 **Failed cards get an inline "Retry →"** — dead-ends always offer the next action on the card itself.

INTERACTIONS

Whole card = tap target (56px+). Long-press opens a bottom action sheet: Open, Download .docx, Delete (destructive red, confirm sheet — replaces the browser-native `confirm()` used today). Pull-to-refresh re-fetches the list; React Query's cache makes it instant.

Spec reader — completed project

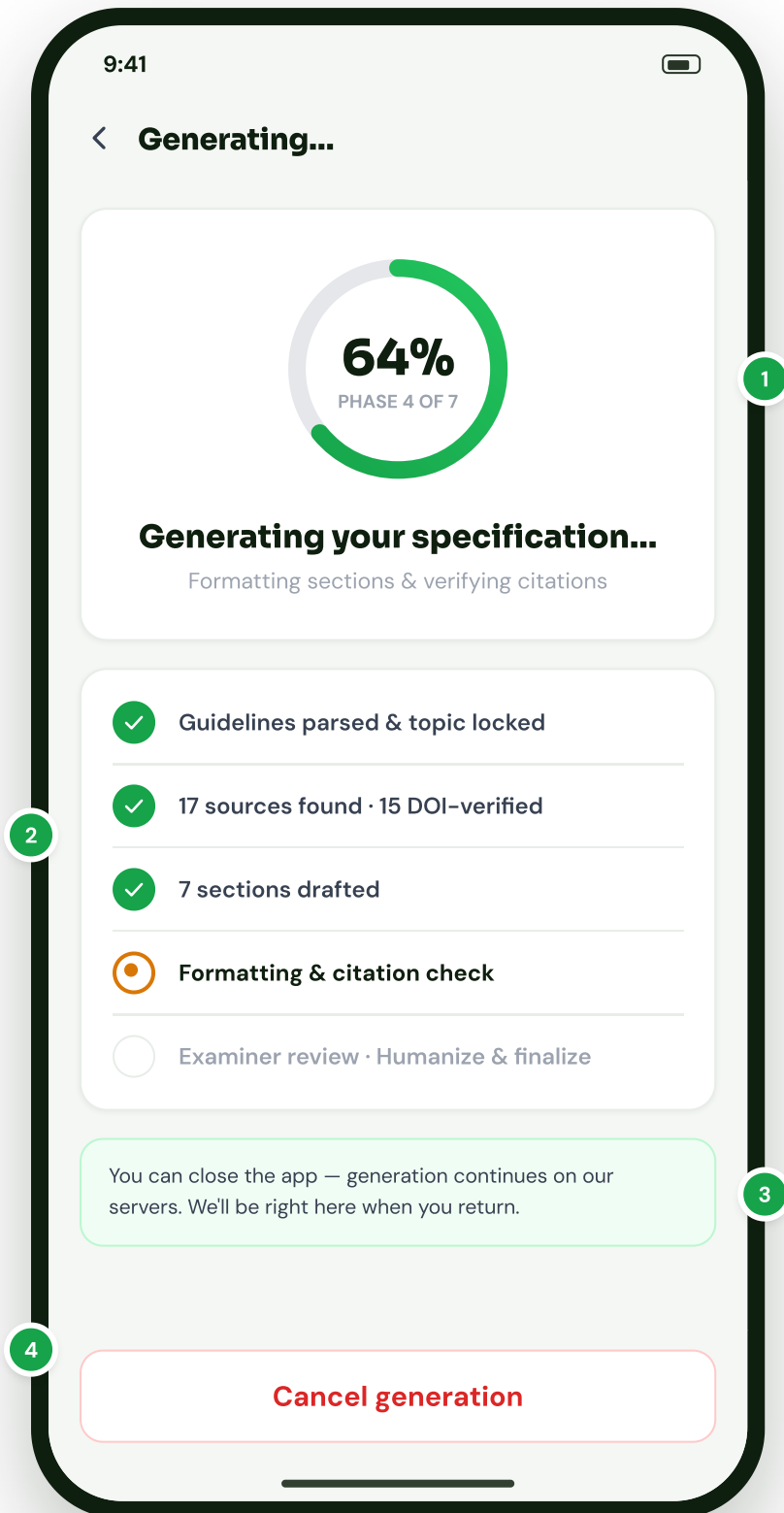


- 1 **Title truncates, status + mark stay pinned** in the app bar — you always know what you're reading and how it scored.
- 2 **Horizontally scrollable section pills** jump-scroll to any of the 7 sections — the mobile stand-in for a desktop table of contents.
- 3 **Score ring for the headline number**, single green (magnitude = one hue), value in ink inside it; the reviewer's overall assessment sits beside it, not below the fold.
- 4 **Section scores as labeled bars** — value labels in ink on every bar (direct labels, no legend needed), single-hue green fills on gray tracks.
- 5 **Download is sticky.** The single most valuable output of the entire product never scrolls away.

READING MODE

Sections are collapsible cards; expanded prose reads in Georgia/Fraunces at 1.75 line-height on white — document typography, distinct from UI chrome. References render as a numbered list with DOI links (CrossRef-verified ones get a small ✓).

Live generation state

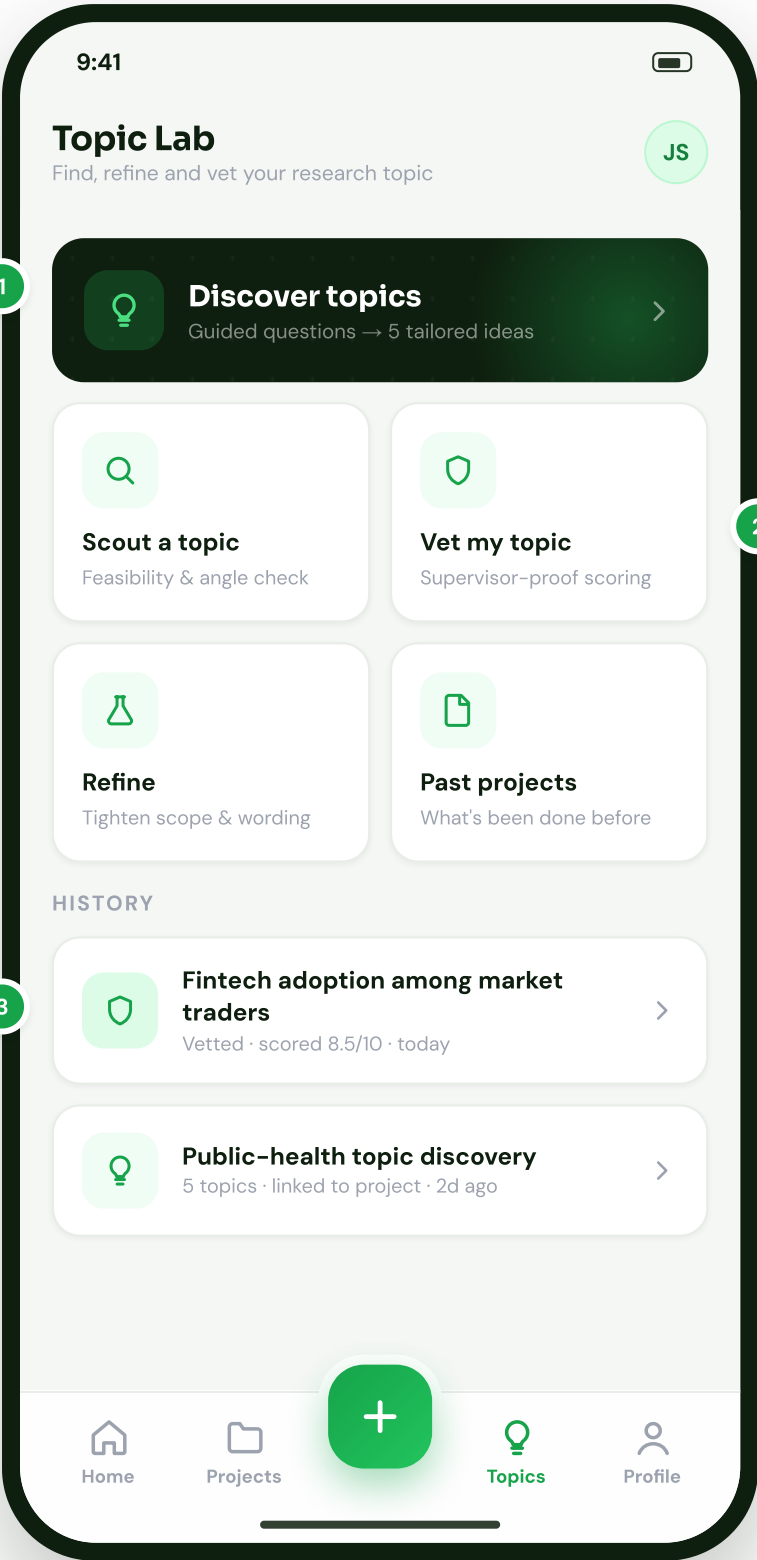


- 1 One big ring, one big number.** The 3-second polling loop the app already runs feeds a 120px progress ring with the phase count under the percentage — status at arm's length, literally.
- 2 Phase checklist with real telemetry** — "17 sources found · 15 DOI-verified" comes straight from the pipeline's own progress messages. Done = green check, active = pulsing amber, pending = hollow. Never color alone.
- 3 "You can close the app"** — the single most important sentence on this screen. Mobile users *will* background the app during a 10-minute generation; telling them it's safe prevents both anxiety and rage-cancels.
- 4 Cancel is quiet, red-outlined, and honest** — it now genuinely stops the task (this week's fix), so the button can finally say what it does. Confirmation sheet before firing.

FAILURE STATE

On FAILED: the ring turns into a red-tinted card with the sanitized error, the free-credit note if it was refunded, and two actions — "Try again" (primary) and "Contact support" (mailto). No dead ends.

Topic Lab — hub

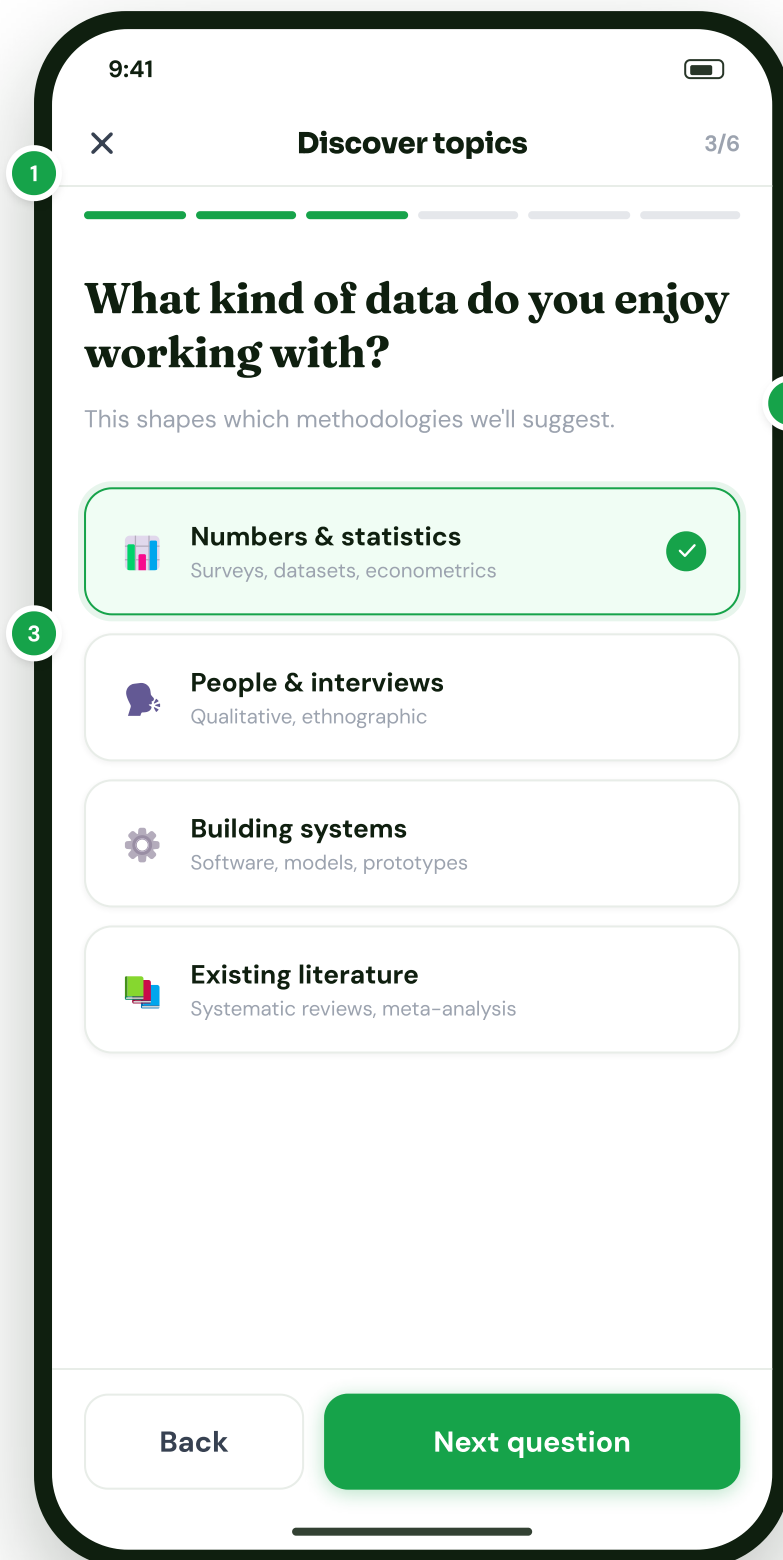


- 1 **Discover gets the hero slot** (dark card — the app's "premium action" pattern) because it's the tool that starts the whole journey for an undecided student.
- 2 **The other four tools as a 2×2 grid** of equal cards — one icon, one job, six words max. The desktop page's 1,160 lines of tabbed UI become five honest entry points.
- 3 **History is first-class**, not a sidebar afterthought: each session row says what kind it was, its outcome ("scored 8.5/10", "linked to project") and when.

STRUCTURAL NOTE

Topic Lab is currently the one surface that bypasses the shared API client and React Query. The mobile rebuild is the natural moment to route it through *lib/api.ts* — the redesign assumes each tool is its own route (*/topics/discover*, */topics/vet...*) instead of one monolithic page, which also fixes back-button behavior.

Guided flow — one question per screen

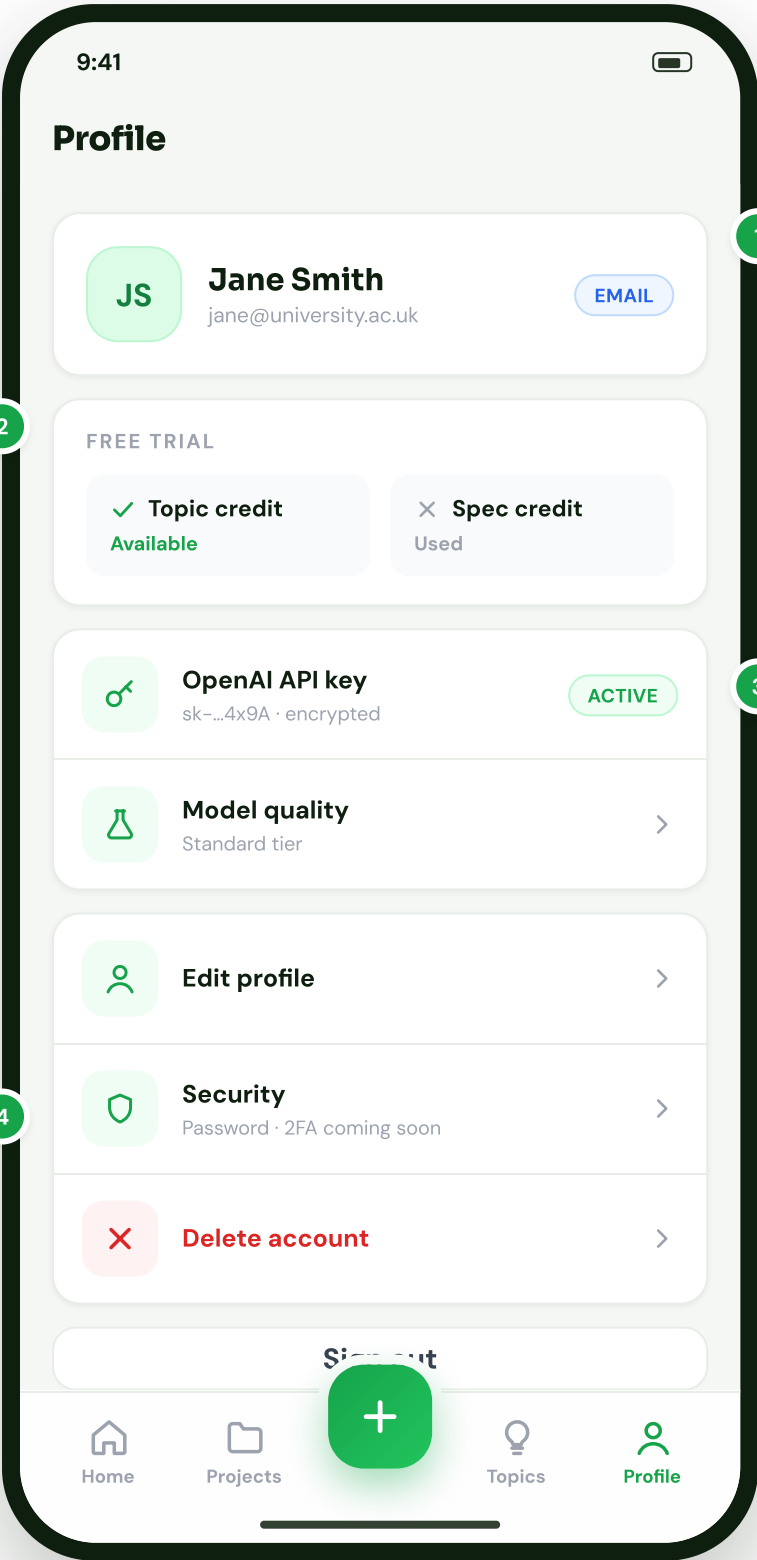


- 1 **X to exit, count to orient.** A guided flow is a modal journey: close is explicit (with a "save your progress?" sheet), progress is honest (3/6).
- 2 **The question is the headline** — Fraunces serif, 1.35rem. The desktop version already uses `QUESTIONS[q].label` as an h2; mobile gives it the whole screen.
- 3 **Answer cards, not radio dots:** emoji + label + example line, whole card tappable, selected state = green border + tinted fill + check. One tap, thumb-sized, self-explanatory.

RESULTS SCREEN

The "Generating your topics..." wait state reuses the progress-ring pattern from sheet 10 (shorter, ~30s). Results arrive as swipeable full-width topic cards — title, cluster badge, complexity, one-line rationale — with "Vet this topic" and "Start a spec from this" actions on each, chaining the tools together.

Profile & settings



- 1 **Provider badge shows how this account signs in** — EMAIL or GOOGLE, driven by the `auth_provider` field added this week. No more hardcoded "Google OAuth 2.0" copy for password users.
- 2 **Trial credits as two honest cells** — Available/Used with icon + word, never color alone. This replaces the sidebar meter, which has no home on mobile.
- 3 **BYOK front and center:** masked key (sk-...4x9A, exactly what the API returns), "encrypted" stated plainly, ACTIVE badge. Tapping opens a sheet to replace or remove the key with the live-validation flow that already exists.
- 4 **Delete account is inside a group, red, and last** — findable but not a landmine. It opens a type-to-confirm sheet ("type DELETE") — the confirmation step the API currently lacks, enforced in UI first.

GROUPED-LIST PATTERN

Settings-style stacked cards with 56px rows — the pattern every phone user already knows from iOS/Android settings. Nothing here needs invention; it needs familiarity.

How to ship this incrementally

ORDER OF WORK

1 • Tab bar + app bar shell ~1 day

One new component pair in *dashboard/layout.tsx* behind the existing 860px breakpoint. The drawer and hamburger are deleted, not maintained in parallel. Desktop keeps the sidebar untouched.

2 • Dashboard, Projects, Profile ~2–3 days

Mostly CSS restructuring: single column, cards instead of tables, stat tiles. The data hooks (*useProjects*, *useUser*) are reused as-is. Consolidate the three duplicated status-color functions into one helper while touching all three pages.

3 • Generate wizard + spec reader ~3–4 days

Real component work: step state for the wizard, accordion + pills for the reader, progress ring for the live state. The polling infrastructure already exists; only presentation changes.

4 • Topic Lab split ~1 week

The big one: break the 1,160-line page into per-tool routes, moving its raw-axios calls onto *lib/api.ts* + React Query in the same pass (fixing the known 401-handling gap). Do it last; everything else ships without it.

GROUND RULES

- **No new design language.** Every screen uses the existing tokens (#16a34a green, #0f1f0f ink, existing badge/card/button classes). This proposal adds layout patterns, not a theme.
- **No new dependencies** except optionally *vaul* (bottom sheets). Icons come from *lucide-react*, already installed — retiring the ~90 hand-rolled SVGs as pages are touched.
- **Accessibility is built in, not bolted on:** every tap target is a real `<button>/<a>`, 44px minimum, with focus-visible states — fixing the current div-onClick pattern as each screen is rebuilt.
- **Browser-native `alert()`/`confirm()` are replaced** by bottom sheets as each destructive action is touched.

WHAT THIS BUYS BEYOND MOBILE

The wizard steps, action sheets, status helper and typed section components are all reusable on desktop — this is the first concrete step toward collapsing the four coexisting design systems into one.

Companion document

The full-frontend aesthetic redesign (desktop) is proposed separately in *frontend-redesign-proposal.pdf* — same tokens, same components, one unified system across both form factors.